

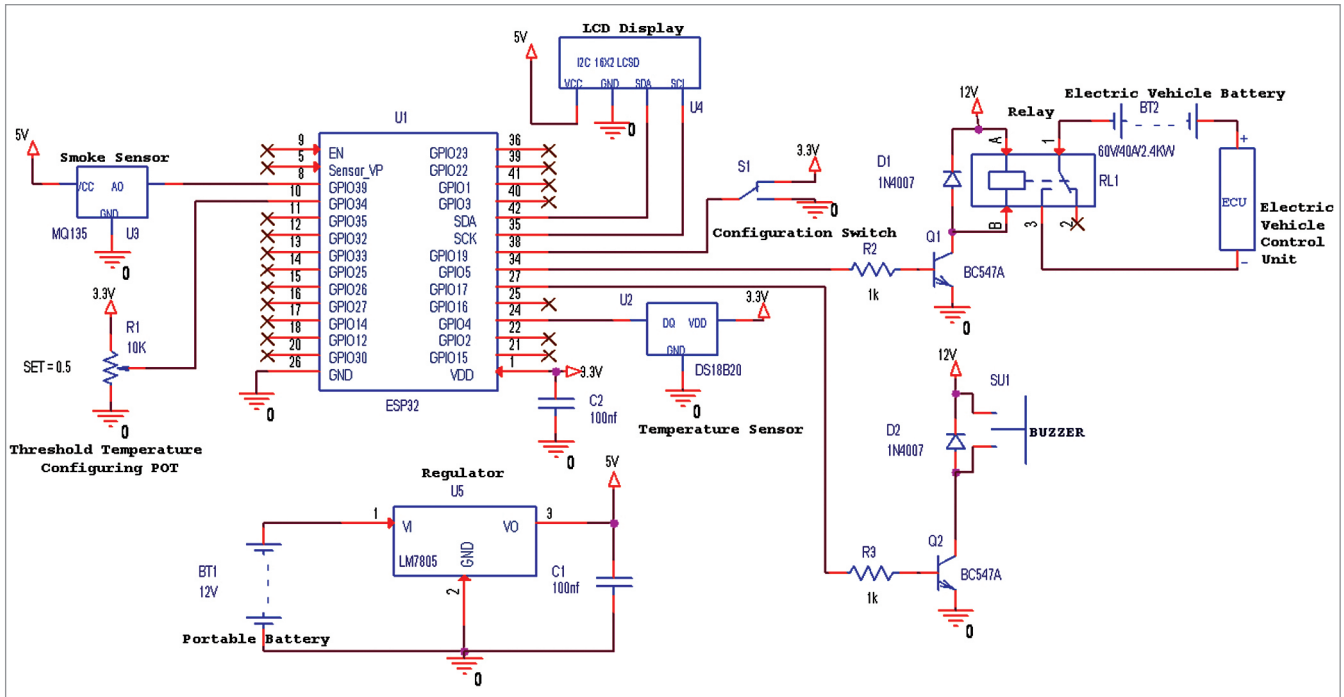


Fig. 3: Circuit diagram of thermal management system

```

tms

#include <OneWire.h>
#include <DallasTemperature.h>
#include <LiquidCrystal_I2C.h>
#include "BluetoothSerial.h"

#if !defined(CONFIG_BT_ENABLED) || !defined(CONFIG_BLUEDROID_ENABLED)
#error Bluetooth is not enabled! Please run 'make menuconfig' to enable it
#endif

BluetoothSerial SerialBT;

String message = "";
char incomingChar;
String temperatureString = "";
String humidityString = "";

#define ADC_VREF_mV 4755.0
#define ADC_RESOLUTION 4096.0
#define MQ135 39
#define TEMP_REF 34

const int oneWireBus = 4;

LiquidCrystal_I2C lcd(0x27, 16, 2);
const int SW = 19;
const int BUZZAR = 17;
const int EV_ECU = 5;
const float threshold2=400.00;

```

Fig. 4: Code snippet

ESP32 board (U1), temperature sensor DS18B20 (U2), smoke sensor MQ135 (U3), 16 × 2 LCD display (U4), 5V regulator LM7805 (U5), and a few other components. The SPDT switch and potentiometer

are connected to input of ESP32 controller while I2C, 16 × 2 LCD display, buzzer, relay, and Arduino Blue Control application are connected to its output. SPDT switch S1 is for mode selection to display

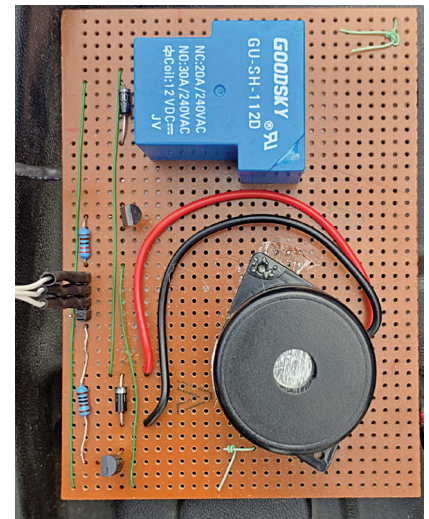


Fig. 5: Relay connection prototype

the current temperature or to configure the threshold.

Software

The Arduino IDE is not ready to program ESP board by default, so go to Board Manager and install it. Install the latest Arduino IDE library for the software. Later install the libraries needed for interfacing the sensor.